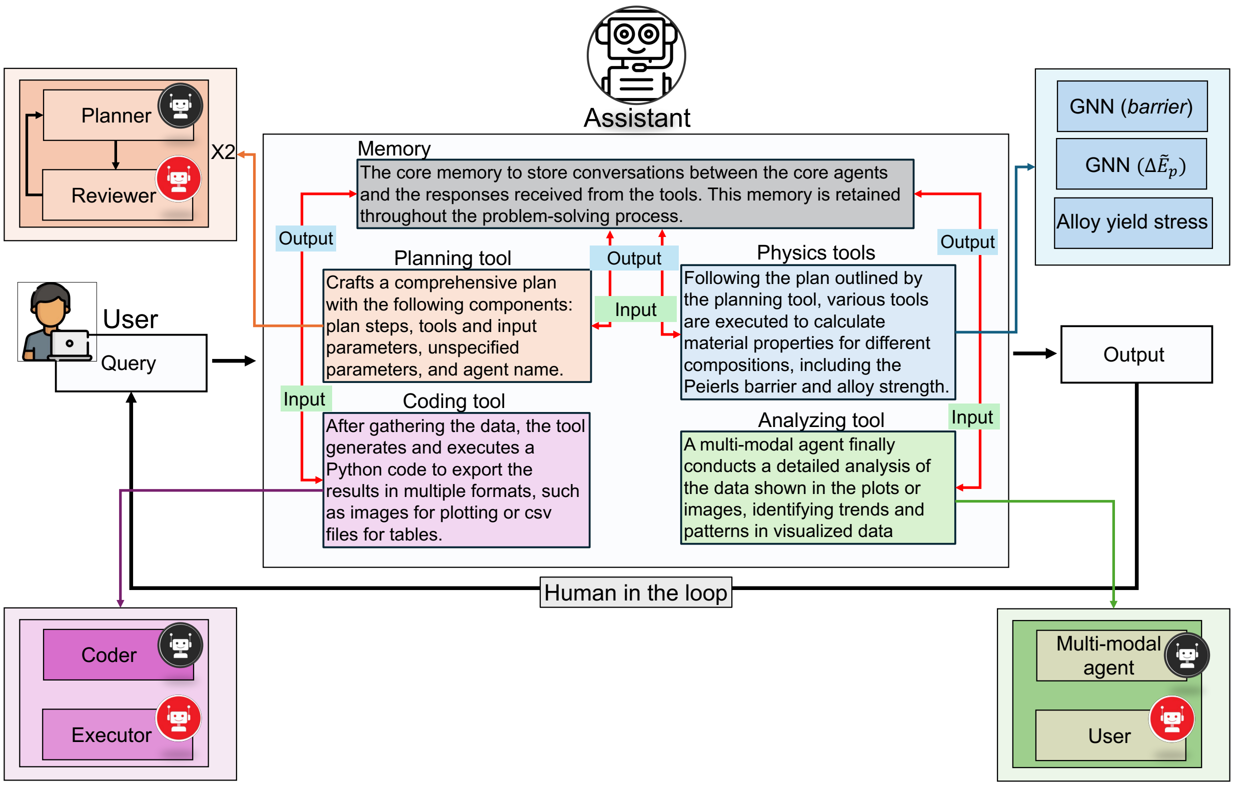

Figure 5: **Overview of a typical workflow executed by our multi-agent system for accelerated and automated multi-component alloy design and analysis.** Upon receiving a query from the user, the process begins with the assistant agent, which calls the planning tool to generate a detailed plan. The assistant agent follows this plan by invoking the relevant tools and providing the necessary input functions. These tools are integrated into our system to equip the agent with advanced capabilities, including material property predictions through physics-based retrieval tools (GNN models), data visualization via code execution (through a code-executor pair of agents), and multi-modal analysis (through a multi-modal agent-user pair) for enhanced visualization and reasoning. The system incorporates a human-in-the-loop design, fostering dynamic collaboration between humans and AI. This enables continuous refinement of ideas and queries, resulting in a more adaptive and efficient problem-solving process.

The Assistant agent then executes the plan, generating the possible compositions in the ternary space and activating the relevant tools, as shown in Figure 6(b). We observe that the GNN model enables rapid predictions of the Peierls barrier across the entire compositional space (231 compositions at 5% intervals). Notably, calculating the mean Peierls barrier for a single composition typically requires computationally expensive NEB simulations over numerous random configurations to capture the statistical nature of the problem. The GNN-based approach, however, completes these calculations within seconds per composition, demonstrating significant efficiency. After computing the Peierls barriers, the coding tool is called to write a Python function for plotting the values on a ternary diagram. Despite the complexity of ternary plots, the code generator, powered by the o1-mini LLM, produces the plot, as shown in Figure 6(c) , depicting the variation of the Peierls barrier across Nb-Mo-Ta ternary and binary compositions.

The results of the multi-agent system provide valuable insights into the variation of the Peierls barrier across the entire compositional space. Regions with high Peierls barrier values are identified around compositions with intermediate levels of Nb and Mo and low levels of Ta. As the final step of the workflow, the multi-modal agent performs detailed analysis, identifying the highest Peierls barrier in compositions with a high percentage of Mo and the lowest in those with a high percentage of Ta. Additionally, the agent conducts a comparative analysis of the influence of Nb, Mo, and Ta concentrations on the Peierls barrier.

However, some inconsistencies are observed in the agent's analysis. For example, the model suggests that the Peierls barrier decreases with increasing Nb content, which is inaccurate based on the predictions. This highlights challenges in the multi-agent system's ability to detect patterns in complex ternary plots, requiring